KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2017
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

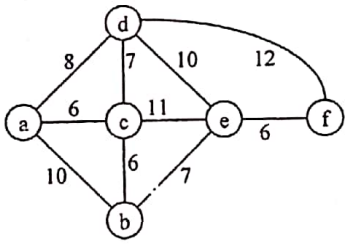
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define statistical multiplexing. Which one is better between statistical multiplexing and Time Division Multiplexing? Explain with an appropriate example. (12)
   b) Suppose source and destination are connected through a switch and a router. If source sends a message to destination then how do encapsulation and decapsulation happen in each layer of TCP/IP model of above mentioned devices? Explain with necessary figure. (11)
   c) Suppose A is transmitting a packet to B. They are directly connected through a physical link. The length of the link is 1 km, the length of the packet is L bits and the transmission rate is 1 Mbps. (i) Now calculate how many bits will be arrived at B in 50 μsec? (ii) If the packet length is 100 bytes then at what transmission rate R, the propagation delay would be equal to the transmission delay of packet? (12)

2. a) Consider sending a 2400 byte datagram into a link that has an MTU of 700 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation? (10)
   b) Suppose the CSE department of KUET has 200 personal computers. The network administrator does not want to set the IP addresses statically. So, how the personal computers can get IP addresses? Explain steps of getting IP. (11)
   c) Consider the following network. With the indicated link costs, use Dijkstra's shortest path algorithm to compute the shortest path from *b* to all network nodes. Show how the algorithm works by computing a table. (14)



3. a) How does private IP addressing concepts resolve IPv4 address space problem? What are the new problems arises? Explain. (09)
   b) Suppose A wants to send a packet to B. If A does not know B's MAC address then how does it determine this MAC address in the following two situations: (13)
      i) A and B are in the same network.
      ii) A and B are in the different network.
   c) What is Ethernet? Describe the working principle of Ethernet's CSMA/CD protocol with a detail look on exponential back-off algorithm. (13)

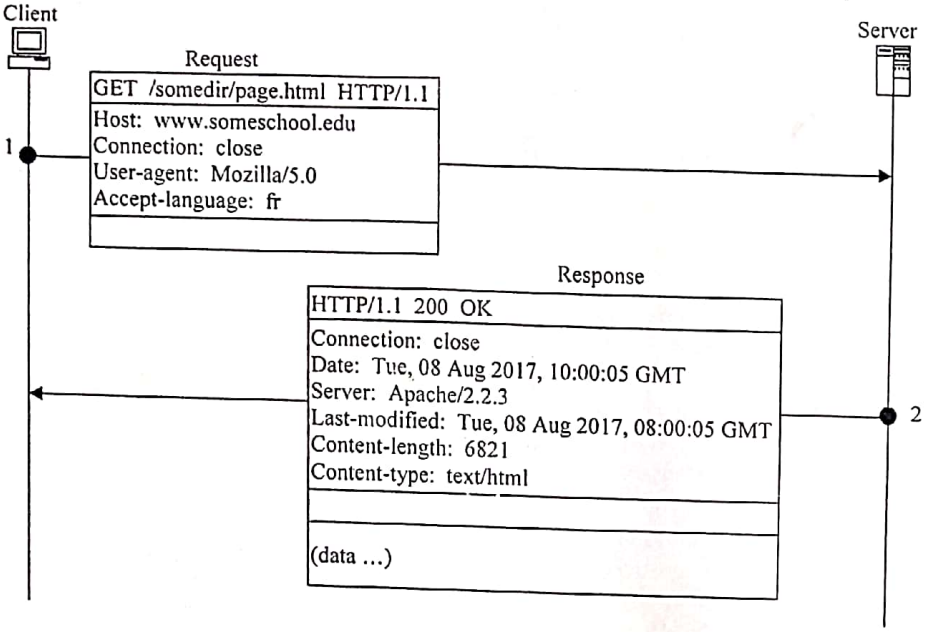
4. a) Describe the technologies that are used for transitioning from IPv4 to IPv6. (06)
   b) How does Multiple Access with Collision Avoidance (MACA) solve hidden terminal problem in wireless network? Explain with example. (08)
   c) How does IEEE 802.11 deal with data fragmentation? Discuss. (10)
   d) The CSE department of KUET just bought a Cisco switch which has 50 ports. The network administrator connected 40 personal computers to that switch. Now explain the learning mechanism of it. (11)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

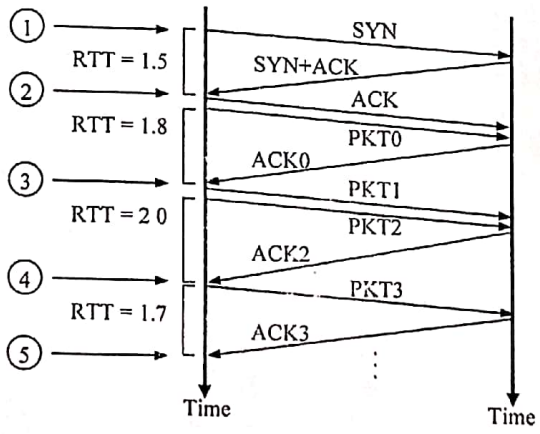
5. a) Compare among the Internet, WWW and HTTP. (09)
   b) What are the limitations of SMTP? How can these be solved by Multipurpose Internet Mail Extensions (MIME)? Briefly explain the working process and *Base64* encoding of MIME. (11)

c) Consider the following communication between a client and a server. The figure shows the different messages transferred between them. Explain each section as well as the statements that are presented in the messages. Also provide a general format of those messages. (15)



6. a) Mention different components of Domain Name System (DNS). Explain various types of name server used in DNS by presenting their architecture and working mechanism. (11)
   b) Design a stop-n-wait type reliable data transfer protocol by showing a finite state machine. Also present the line diagrams to explain its operation for various packet transmission scenarios. (14)
   c) How does selective repeat protocol work? Explain. (10)

7. a) How does TCP recognize the network congestion? Explain, with necessary illustration, the Reno approach of managing this congestion. (14)
   b) What is silly window syndrome? How does TCP handle this issue? (08)
   c) How does TCP set the timeout interval for retransmission timer? What are the Jacobson/Karels proposals for setting that interval? Consider the following communication scenario of TCP. Figure shows the sample RTTs in seconds. Compute the timeout intervals at the points shown by arrows i.e. at initial position and at the end of each acknowledge received. (13)



8. a) Design a protocol to authenticate a live entity. Can it be used for public key system? Why or why not? (12)
   b) Let you are a network administrator, design a rule set for your packet filter firewall to allow your local users only to browse outside and no other traffic except the locally initiated return traffic from outside is allowed to enter your network. (07)
   c) Briefly explain how Secure Socket Layer (SSL) works on top of TCP and what services it provides. (09)
   d) What are the basic differences between TCP and UDP? (07)